EC2200

Economics of Managerial Decision Making

Winter 2023 Examination · Paper 1

Duration 1.5 hours · Section A: 2 of 3 short questions (50%) · Section B: 1 of 2 long questions (50%)

Module	EC2200 — Economics of Managerial Decision Making
Session	Winter 2023
Structure	Section A: three short questions, answer any two (each marked out of its own 100%, contributing 50% of the paper between them). Section B: two long questions, answer one (marked out of 100%, contributing the remaining 50%).
This scheme covers	All three Section A questions and both Section B questions in full.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the marks for the later steps done correctly on your own figures.

How this paper relates to the other EC2200 papers. This paper introduces the Pizza Hut/Ford deficient-demand case study and the Intel-style perfectly competitive industry problem, both distinct to this sitting. The decision-making process, shareholder wealth-maximisation, Herfindahl Index, price discrimination framework, and Porter's Five Forces are shared building blocks with consistent underlying content across the EC2200 papers in this set — only the numbers and framing differ here.

Section A · Question 1

Decision-Making, Profit Maximisation & Shareholder Wealth · 40+10+10+40%

A1(a) **40%**

"The ability to make good decisions is the key to successful managerial performance". Describe and illustrate each step in the decision-making process, clearly labeling each stage in the process. Support your answer with a relevant example.

A1(b) **20%**

Suppose the profit, π, of a firm is represented as a function of the output level, Q, using the expression $\pi = -40 + 140Q - 10Q^2$.
(i) Calculate the profit maximizing level of output for the firm.
(ii) Calculate the firm's profit at the profit maximizing level of output.

A1(c) **40%**

Explain how the shareholder wealth-maximization model overcomes the limitations of the simple profit-maximation model.

(a) The decision-making process — 40%

The Managerial Decision-Making Process

Good managerial decisions follow a structured process rather than ad-hoc judgement. The standard steps, illustrated with a worked example (a firm deciding whether to launch a new product line):

- **Step 1 — Define the problem/objective.** State clearly what is being decided and against what criterion it will be judged (usually profit or firm value maximisation). *Example: should the firm launch Product X, judged against its effect on total profit?*
- **Step 2 — Identify the constraints.** The technology, budget, production capacity, regulatory environment, and competitive conditions that limit which choices are actually feasible. *Example: the firm has a fixed production line capacity and a maximum marketing budget for the launch.*
- **Step 3 — Identify the alternatives.** The realistic set of options available given the constraints — e.g. launch now, delay and gather more market research, launch a scaled-down version, or don't launch at all.
- **Step 4 — Evaluate the alternatives at the margin.** For each alternative, compare the marginal (incremental) revenue it would generate against the marginal (incremental) cost it would require — not average or total figures, which can mislead. *Example: compare the additional expected revenue from launching against the additional cost of production and marketing.*
- **Step 5 — Choose the best alternative and implement it.** Select the option that maximises the objective from Step 1, subject to the constraints from Step 2, then put it into action.
- **Step 6 — Monitor outcomes and feed back.** Track the actual results against the decision's expected outcomes, and use this information to inform the next decision (returning to Step 1) — making the process a continuous cycle rather than a one-off exercise.

> **Why marginal analysis (Step 4) is the crux of the process:** a decision should be taken if, and only if, its marginal benefit exceeds its marginal cost — comparing totals or averages can lead to the wrong decision (e.g. a profitable overall product line might still contain an unprofitable marginal unit). This is the same principle underlying the MR=MC profit-maximisation rule used throughout this course.

(b) Profit-maximising output — 10% + 10%

Step	Working
Profit function	$\pi = -40 + 140Q - 10Q^2$
First derivative	$d\pi/dQ = 140 - 20Q$
(i) Set equal to zero and solve	**$140 - 20Q = 0 \Rightarrow Q^* = 7$**
Second derivative (check maximum)	$d^2\pi/dQ^2 = -20 < 0 \Rightarrow$ confirmed maximum
(ii) Substitute Q*=7 into π	$\pi^* = -40 + 140(7) - 10(7)^2 = -40 + 980 - 490$
Profit-maximising output and profit	**$Q^* = 7$, $\pi^* =$ €450**

(c) How shareholder wealth-maximisation overcomes profit-maximisation's limitations — 40%

Main features of the shareholder wealth-maximisation model

> **The objective:** maximise the present value (PV) of the firm's expected future profits (or dividends per share), discounted at the shareholders' required rate of return — not simply maximise this period's accounting profit.

$$V_0 = \Sigma\, \pi_t / (1+r)^t$$

> **It is forward-looking and multi-period:** the model explicitly values profits expected in every future period, not just the current one, so a decision that sacrifices some profit today for materially higher profit later can still raise firm value.

> **It incorporates the time value of money:** profits expected further in the future are discounted more heavily (at rate r, compounded over t periods) — a euro of profit next year is worth less today than a euro of profit now.

> **It incorporates risk via the discount rate:** the required rate of return (r) reflects the riskiness of the firm's expected profit stream — riskier, more uncertain profit streams are discounted at a higher rate, reducing their contribution to firm value, all else equal.

How this overcomes the limitations of the simple profit-maximisation model

Limitation of simple profit maximisation	How shareholder wealth-maximisation resolves it
Focuses on a single period's profit only	Values the entire expected future stream of profits, not just the current period — discourages short-termism (e.g. cutting valuable long-run investment just to boost this quarter's profit).
Ignores the timing of returns	Explicitly discounts profits by how far in the future they arrive, via the $(1+r)^t$ term — a strategy that shifts profit earlier in time is recognised as more valuable, all else equal.
Ignores risk	The discount rate r is set according to the riskiness of the firm's expected profit stream — riskier strategies must promise a higher expected return to be judged equally attractive, correctly penalising excessive risk-taking that simple profit maximisation would ignore.
Can encourage myopic decision-making	Because value depends on the entire discounted stream, sacrificing near-term profit for a sufficiently large increase in longer-run profit can raise V_0 even though it lowers this period's accounting profit — aligning decisions with long-run firm value rather than short-run appearances.

The key generalisation: simple profit maximisation is really a special (single-period, riskless, undiscounted) case of the shareholder wealth-maximisation model — the SWM model doesn't replace the marginal-analysis logic (MR=MC) used to find the profit-maximising output in any given period, it just applies that logic within a framework that correctly weighs timing and risk across every period, not just one.

Mark Allocation — Part (a) (40%)

Tier	Marks	What separates this tier
32–40%	All steps of the decision-making process correctly described and illustrated, with a relevant worked example.	
20–31%	Most steps described correctly but the illustration or example is thin.	
0–19%	Partial or generic description without a clear step-by-step process.	

Mark Allocation — Part (b) (10% + 10%)

Tier	Marks	What separates this tier
Full marks	Correct first derivative, $Q^*=7$ correctly solved, second-order condition checked, and $\pi^*=€450$ correctly calculated.	
Partial	Correct Q^* but the second-order condition omitted, or an arithmetic error in computing π^* carried through consistently.	
Minimal	Incorrect derivative or incorrect solving.	

Mark Allocation — Part (c) (40%)

Tier	Marks	What separates this tier
32–40%	Correctly identifies at least three limitations of simple profit maximisation and explains how the SWM model's PV/discounting framework resolves each.	
20–31%	General discussion of SWM being 'more comprehensive' without specific limitation-by-limitation resolution.	
0–19%	Minimal or incorrect treatment.	

Question A1 Total: 100%

Section A · Question 2

Deficient Demand, Price Elasticity & Revenue Maximisation · 40+30+30%

A2 — Case Study

CASE STUDY: Pizza Hut and FORD Dealers Respond to Deficient Demand

Pizza Hut anticipates a purchase frequency of 60 pizzas per night at a price of \$9 and plans their operations accordingly. When less than 60 customers arrive on a given evening, **the Pizza Hut franchisee does something different from what a FORD auto dealer might do in similar circumstances**. The restaurant slashes orders; fewer pizza dough balls are flattened and spun out and baked. Instead of slashing prices in the face of deficient demand, restaurants tend to delay their release-to-production orders.

A2(a)

40%

Based on the CASE STUDY: Explain how a FORD auto dealer might react in similar circumstances to deficient demand and why the two businesses react differently.

A2(b)

30%

Automobile manufacturers have discovered that a temporary price cut raises more revenue than a permanent price reduction. Explain in terms of their demand and supply elasticity.

A2(c)

30%

Demonstrate for the demand function $Q_d = 100 - P$, that marginal revenue (MR) = 0, and Total Revenue (TR) is maximized at unit elasticity, where Q_d = quantity demanded, and P = Price.

FULL MODEL ANSWER

(a) Why Ford and Pizza Hut react differently to deficient demand — 40%

Ford's likely response: cut price (offer rebates/incentives). cars are durable, storable goods with a high per-unit value and significant holding costs (financing/interest cost of unsold inventory sitting on a dealer's lot, obsolescence as newer model years arrive, lot space). An unsold car today can still be sold tomorrow, but every day it sits unsold is a real, measurable carrying cost to the dealer. Cutting price (or offering a rebate/incentive) is a standard, well-understood way to clear excess inventory quickly and stop those holding costs accumulating, and the auto industry has established, expected mechanisms (rebates, 0% financing offers, dealer incentives) for doing exactly this without permanently repricing the base sticker price.

Pizza Hut's actual response: cut quantity (delay production), not price. a pizza is a perishable good produced to near-immediate order, with a low marginal cost of production relative to price and effectively zero salvage value if unsold (unlike a car, an un-made pizza dough ball simply isn't made at all — there's no 'unsold inventory' sitting around incurring holding costs). Cutting price to fill empty tables offers little benefit: the marginal cost saved by not making a pizza is small, so there's little lost by simply making fewer; meanwhile, a price cut risks setting a new reference price in customers' minds (once customers see pizza discounted, they may delay future full-price purchases waiting for the next discount — a demand-side risk with no equivalent inventory-cost benefit to offset it for a good this cheap to simply not produce).

The underlying economic distinction: the choice between adjusting price versus adjusting quantity in response to a demand shortfall depends on the relative cost of holding unsold inventory versus the cost (including reputational/reference-price cost) of cutting price. Durable, storable, high-value goods with real holding costs (cars) favour price adjustment; perishable, low-marginal-cost goods with near-zero cost of simply not producing the marginal unit (pizza) favour quantity adjustment instead.

(b) Why a temporary price cut raises more revenue than a permanent one — 30%

Temporary discounts exploit high short-run (intertemporal) elasticity of demand. when a price cut is known to be temporary, price-sensitive consumers who might otherwise have purchased later bring their purchase forward in time to take advantage of the limited-time deal — a large, concentrated surge in quantity demanded during the discount window, reflecting high elasticity of demand with respect to the timing of the discount specifically.

A permanent price cut doesn't generate this same surge, and sacrifices margin on every future unit. because a permanent cut doesn't create urgency (there's no reason to buy today rather than next month, since the lower price will still be there), it doesn't pull forward future demand into the present the way a temporary discount does. Worse, a permanent cut locks in the lower price on all future sales, including sales that would have happened anyway at the original price — sacrificing revenue on inframarginal sales that a temporary cut, by construction, does not sacrifice once the promotional window ends and the price reverts.

Reference-price effects reinforce this: a temporary, clearly-time-limited discount is understood by consumers as an exception, preserving the 'normal' price as the reference point for future purchases; a permanent price cut resets that reference point downward, permanently reducing the price consumers are willing to pay going forward and making future price rises harder to achieve without a demand backlash.

(c) MR = 0 and TR maximised at unit elasticity — 30%

Step	Working
Demand function	$Q_d = 100 - P \Rightarrow P = 100 - Q$
Total revenue	$TR = P \times Q = 100Q - Q^2$
Marginal revenue	$\mathbf{MR = dTR/dQ = 100 - 2Q}$
Set MR = 0	$100 - 2Q = 0 \Rightarrow Q^* = 50, P^* = 50$

Confirming unit elasticity: $E_d = (dQ/dP) \times (P/Q) = -1 \times (50/50) = \mathbf{-1}$ at $Q^*=50$ — exactly unit elastic, confirming TR is maximised exactly where MR=0 and demand is unit elastic, the same general result demonstrated on the equivalent question elsewhere in this set of papers.

Mark Allocation — Part (a) (40%)

Tier	Marks	What separates this tier
32–40%	Correctly identifies Ford's likely price-cutting response and Pizza Hut's quantity-adjustment response, and explains the underlying economic distinction (holding costs of durable vs. perishable goods, marginal cost of the unsold unit, reference-price risk).	
20–31%	Correctly identifies the different responses but the underlying economic reasoning is thin or one-sided.	
0–19%	Vague or largely incorrect treatment.	

Mark Allocation — Part (b) (30%)

Tier	Marks	What separates this tier
24–30%	Correctly explains the intertemporal demand elasticity mechanism (pulling forward future demand) and the margin-sacrifice/reference-price cost of a permanent cut.	
14–23%	Partial explanation covering one of the two mechanisms.	
0–13%	Minimal or incorrect treatment.	

Mark Allocation — Part (c) (30%)

Tier	Marks	What separates this tier
24–30%	Correct derivation of TR/MR, correctly solves MR=0 for Q*=50,P*=50, and correctly demonstrates unit elasticity at that point.	
14–23%	Correctly finds Q*=50 but the unit elasticity demonstration is missing or incomplete.	
0–13%	Incorrect derivation.	

Question A2 Total: 100%

Section A · Question 3

Optimal Input Use, Elasticity Forecasting & Porter's Five Forces · 20+20+20+40%

A3

Suppose a firm using two inputs, capital (K) and labour (L), has selected the levels of usage of capital (K) and labour (L) at which the marginal product of capital (MP_K) = 60 and the marginal product of labour (MP_L) = 30. If the price of capital = €12 per unit, and the price of labour = €5 per unit:

A3(a)(i) **20%**

What advice would you give to the above firm in terms of optimizing its utilization of labour and capital?

A3(a)(ii) **20%**

Suppose that interest rates declined, resulting in a decrease in the price of capital to €10 per unit, how does this impact on the strategy of the firm in terms of optimizing its utilization of labour and capital.

A3(b) **20%**

The price elasticity of demand facing the Internet is −0.5, the income elasticity of demand is +7.4. Suppose that income is expected to increase 4% this year, and that the price of access to the Internet is anticipated to rise 2%, calculate the percentage change expected from this information for users of the Internet.

A3(c) **40%**

Describe, with the aid of an illustration, the main features of Porter’s Five Forces Strategic Model and discuss how the threat of new entrants is dealt with by Porter.

FULL MODEL ANSWER

(a) Optimising the use of capital and labour — 20% + 20%

The condition for a cost-minimising (optimal) input combination is that the marginal product per euro spent is equalised across all inputs: $MP_K/P_K = MP_L/P_L$. If the ratios are unequal, the firm can get more output for the same total spend by reallocating toward whichever input has the higher ratio.

Input	MP	Price	MP/Price (marginal product per €)
(i) Capital	60	€12	5.0
(i) Labour	30	€5	6.0
(ii) Capital (price falls to €10)	60	€10	6.0
(ii) Labour (unchanged)	30	€5	6.0

(i) Advice at the original prices: labour delivers 6 units of marginal product per euro spent, versus only 5 for capital — labour is currently the more productive euro-for-euro input. The firm should **shift spending toward labour and away from capital** until the two ratios are equalised, which will raise total output for the same total input spend.

(ii) Impact of capital's price falling to €10: the ratios become exactly equal (6.0 for both capital and labour at these marginal product levels) — the firm's input mix is now optimal at this snapshot, and **no further reallocation is needed**. This illustrates directly how a fall in an input's price (e.g. from lower interest rates reducing the cost of capital) shifts the optimal input mix toward using more of that now-cheaper input, consistent with the standard theory of cost-minimising input choice.

Caveat for a complete answer: these marginal products (60 and 30) are given as a fixed snapshot; in reality, as the firm actually shifts toward more labour/less capital, diminishing marginal returns mean MP_L would fall and MP_K would rise, so the true optimum is reached once the ratios equalise *at the new levels of input use*, not necessarily exactly at the reallocation implied by the current snapshot alone.

(b) Combined elasticity effect on Internet usage — 20%

Step	Working
Formula	$\%\Delta Q_d = E_d \times \%\Delta P + E_i \times \%\Delta \text{Income}$
Price effect	$E_d \times \%\Delta P = (-0.5) \times (2\%) = -1.0\%$
Income effect	$E_i \times \%\Delta \text{Income} = (7.4) \times (4\%) = +29.6\%$
Combined effect	**$-1.0\% + 29.6\% = +28.6\%$**

Interpretation: even though the 2% price rise reduces quantity demanded by 1% on its own (a modest effect, since demand is fairly price-inelastic at −0.5), the very high income elasticity of +7.4 (Internet access behaves as a strongly income-sensitive, almost luxury-like good in this scenario) means the 4% income growth alone would raise demand by 29.6% — the income effect completely dominates the price effect, so the **net expected change is a 28.6% increase** in the number of Internet users.

(c) Porter's Five Forces — 40%

Porter's Five Forces Strategic Model

Porter's Five Forces Strategic Model

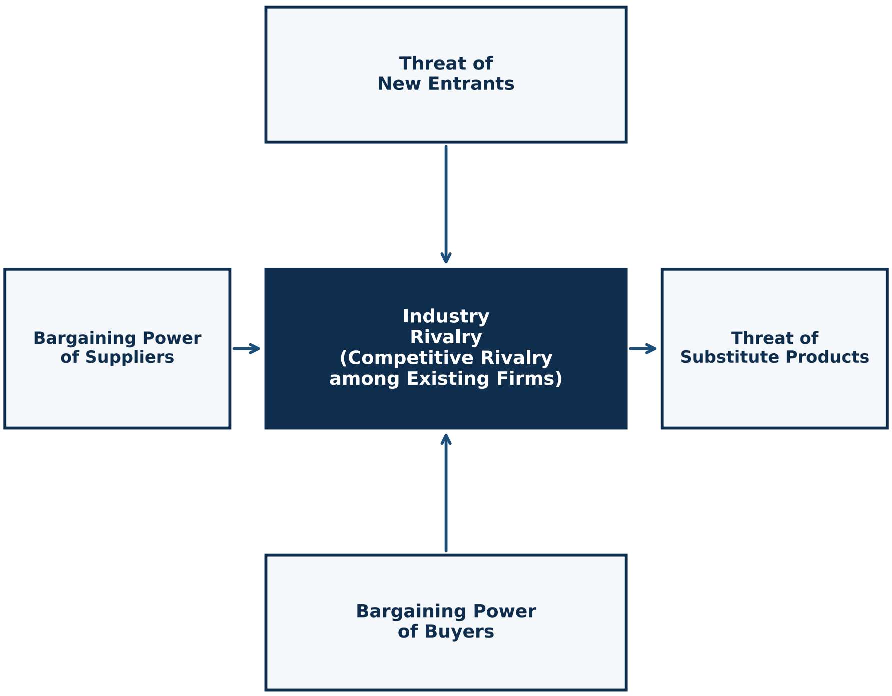

Fig. — The five forces shaping industry competitiveness and profitability

Michael Porter's framework identifies five competitive forces that jointly determine the intensity of competition — and hence the long-run profitability — of an industry. The stronger each force, the more it erodes the potential for firms in the industry to earn above-normal profit.

- **Threat of new entrants:** the ease with which new firms can enter the industry and compete away incumbents' profits — governed largely by barriers to entry (discussed in detail below).
- **Bargaining power of buyers:** the ability of customers to negotiate lower prices or better terms — strong when buyers are large/concentrated, purchases are a large share of buyer spending, products are undifferentiated, or switching suppliers is easy.
- **Bargaining power of suppliers:** the ability of input suppliers to raise prices or reduce quality/terms — strong when suppliers are concentrated, inputs are differentiated or have no substitutes, or switching suppliers is costly.
- **Threat of substitute products:** the availability of alternative products or services that satisfy the same underlying customer need, which caps how much firms in the industry can charge before customers switch away entirely.
- **Industry rivalry (competitive rivalry among existing firms):** the intensity of head-to-head competition among the firms already in the industry — higher when there are many similarly-sized competitors, industry growth is slow, products are undifferentiated, or exit barriers are high.

How Porter addresses the threat of new entrants specifically

Porter argues that the threat of new entrants depends on the height of **barriers to entry** — structural features of the industry that make it costly or difficult for a new firm to enter and compete effectively, even if the industry is currently profitable:

- **Economies of scale:** if incumbents produce at a large enough scale that average costs are much lower than a new, small entrant could achieve, the entrant is forced to choose between entering at a large (risky, capital-intensive) scale or accepting a cost disadvantage.
- **Capital requirements:** industries requiring very large upfront investment (factories, R&D, brand-building) deter entry by all but the best-resourced potential competitors.
- **Product differentiation and brand loyalty:** where incumbents have

established strong brand identity and customer loyalty, a new entrant must spend heavily on marketing simply to become a credible alternative.

- **Switching costs:** if customers face a real cost (financial, time, or effort) in switching from an incumbent to a new entrant's product, the entrant must overcome this inertia, typically by offering large price or quality advantages.
- **Access to distribution channels:** if existing distribution channels (retail shelf space, wholesalers, platforms) are already tied up by incumbents, a new entrant may need to create its own channels — a significant additional cost and barrier.
- **Government policy and regulation:** licensing requirements, patents, and other regulatory barriers can legally restrict entry regardless of a potential entrant's efficiency.
- **Expected retaliation:** if incumbents have a credible history or capacity for aggressive retaliation against new entrants (price wars, capacity expansion), this deters entry even absent any structural barrier, since a rational entrant anticipates the cost of a fight.

> **Porter's overall point:** the threat of new entrants is not simply about whether entry is legally possible, but about how high these structural and strategic barriers are — an industry can be highly profitable yet remain uncontested if barriers to entry are high enough that a rational new entrant expects entry to be unprofitable even against currently high incumbent margins.

Mark Allocation — Part (a) (20% + 20%)

Tier	Marks	What separates this tier
Full marks	Correctly computes MP/Price ratios for both inputs, correctly advises shifting toward labour (i), and correctly identifies that the ratios equalise once capital's price falls to €10, requiring no further reallocation (ii).	
Partial	Correct direction of advice but the MP/Price ratio rule not explicitly used, or (ii) not clearly linked to the new equalised ratios.	
Minimal	Incorrect or no meaningful comparison of the two inputs.	

Mark Allocation — Part (b) (20%)

Tier	Marks	What separates this tier
16–20%	Correct formula, correct calculation of both the price and income effects separately, and the correct combined answer (+28.6%).	
8–15%	Correct method with an arithmetic slip carried through consistently.	
0–7%	Incorrect formula or method.	

Mark Allocation — Part (c) (40%)

Tier	Marks	What separates this tier
30–40%	Correctly describes all five forces with a correct diagram, and gives a thorough discussion of at least four distinct barriers to entry.	
18–29%	Describes the five forces reasonably well but the threat-of-new-entrants discussion is limited.	
0–17%	Partial or generic description.	

Question A3 Total: 100%

Section B: Long Questions

Answer 1 of 2 · this scheme covers both in full

Section B questions are broader and typically integrate several ideas together.

Section B · Question 1

Market Concentration, Price Discrimination & Pricing Structures · 20+10+10+45+15%

B1(a)

30%

Market share distributions for the US Ready-to-Eat cereal market for the top firms are presented for 2018 in percentage terms in Table B1: Kellogg 35%, General Mills 25%, Post 12%, Nabisco 10%, Quaker 7%, Ralston 6%, Others 5%.
(i) Calculate the Herfindahl H-Index for the US Ready-to-Eat cereal market based on the data in Table B1.
(ii) Interpret the Herfindahl H-Index as an indicator of the amount of competition amongst the firms.

B1(b)

10%

Describe the conditions necessary for price discrimination to exist in a market.

B1(c)

45%

Distinguish between first-degree, second-degree and third-degree price discrimination and provide an example of each, clearly illustrating the main economic elements.

B1(d)

15%

Distinguish between bundling and a two-part tariff as methods of price discrimination. Support your answer with an example of each.

FULL MODEL ANSWER

(a) The Herfindahl H-Index for the US cereal market — 20% + 10%

Firm	Market share (%)	Share^2
Kellogg	35	1,225
General Mills	25	625
Post	12	144
Nabisco	10	100
Quaker	7	49
Ralston	6	36
Others	5	25
HHI (sum of squared shares)		**2,204**

(ii) Interpretation: at 2,204, the US ready-to-eat cereal market falls in the **moderately concentrated** range (1,500–2,500) under the standard antitrust thresholds — but notably close to the 2,500 boundary into 'highly concentrated' territory. With Kellogg and General Mills alone accounting for 60% of the market between them, this indicates a market with meaningfully limited competition among a small number of dominant players, even though it falls just short of the 'highly concentrated' classification.

(b) Conditions necessary for price discrimination — 10%

Conditions necessary for price discrimination to exist

- **The firm must have some market/monopoly power.** A price-taking firm in a perfectly competitive market cannot price discriminate — it must be a price-maker with some ability to set price above marginal cost.
- **The firm must be able to identify or segment consumers with different price elasticities of demand.** Price discrimination relies on charging different prices to groups (or units) that differ in how sensitive they are to price.
- **Resale (arbitrage) between segments must be preventable.** If a consumer who buys at the low price could resell to a consumer facing the high price, the price differential would be competed away — effective price discrimination requires segments to be kept separate (e.g. through geography, timing, personal identification, or the nature of the service itself).

(c) The three degrees of price discrimination — 45% (3 × 15%)

First-degree (perfect) price discrimination

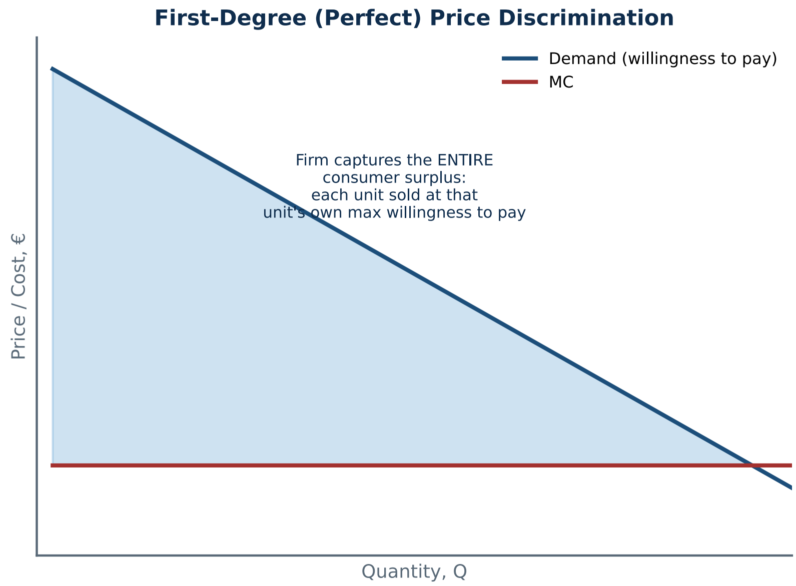

Fig. — First-degree PD: the firm captures the entire consumer surplus

> **What it is:** the firm charges each individual consumer their exact maximum willingness to pay for each unit — extracting the entire consumer surplus as producer surplus, with no deadweight loss (output expands to the competitive/efficient level).

> **Example:** a car salesperson negotiating an individual price with each buyer based on what they read of that buyer's willingness to pay; a personalised/algorithmic online pricing system charging each customer a different price based on their browsing history and data profile.

Second-degree price discrimination

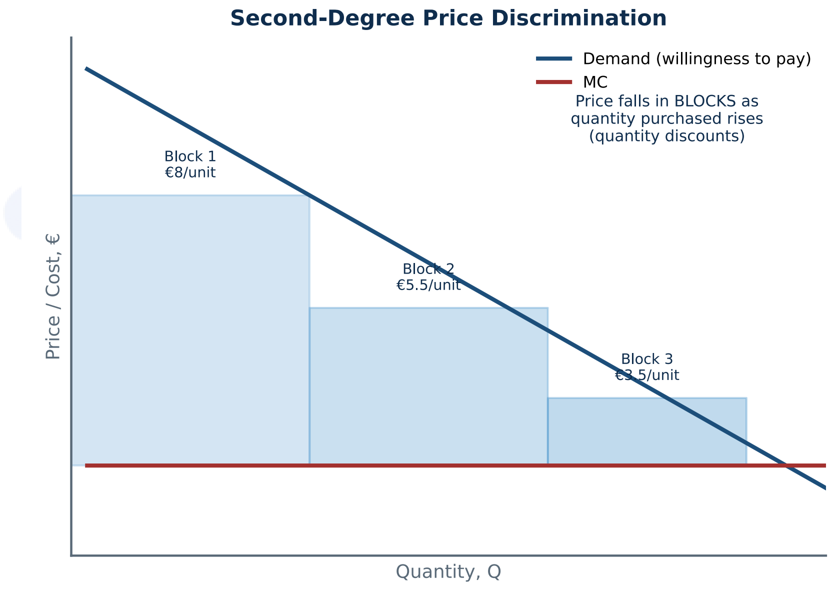

Fig. — Second-degree PD: price varies with the quantity purchased (block pricing)

> **What it is:** the firm charges different prices depending on the quantity a consumer buys (quantity discounts/block pricing), rather than requiring the firm to identify who each individual consumer is — consumers self-select into different blocks based on their own demand.

> **Example:** utility bills charging a lower per-unit rate for electricity/water consumed beyond a certain threshold; bulk-buy discounts (buy 2, get 1 half price); tiered SaaS subscription plans (Basic/Pro/Enterprise) with different quantities/features bundled at different per-unit prices.

Third-degree price discrimination

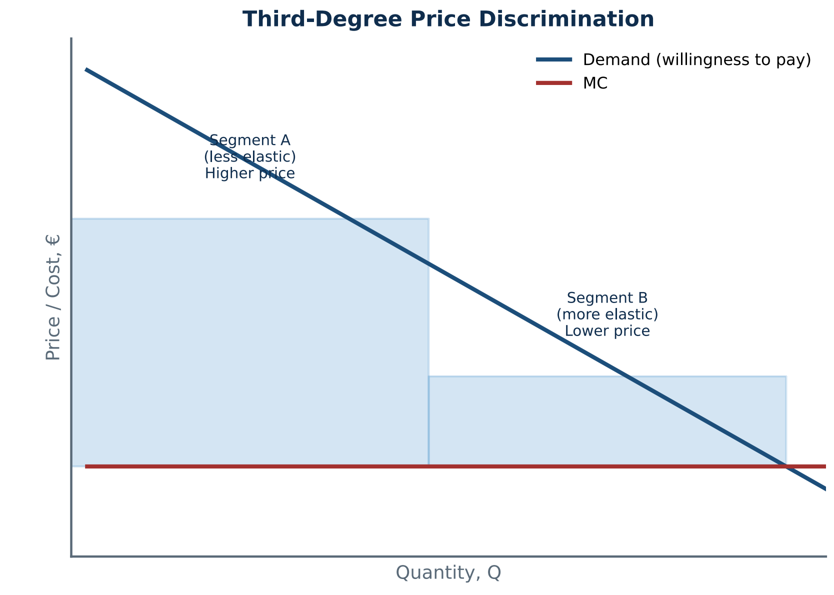

Fig. — Third-degree PD: price varies across identifiable consumer segments by their elasticity of demand

> **What it is:** the firm charges different prices to different, identifiable groups of consumers, based on differences in each group's price elasticity of demand — the group with less elastic (more price-inelastic) demand is charged the higher price, and the group with more elastic demand is charged the lower price (exactly the pattern seen in the Intel Japan/US case, where Japan's less elastic demand was charged the higher price).

> **Example:** student and senior citizen discounts on public transport and cinema tickets; regional/country-specific pricing for the same software or media subscription; off-peak vs. peak pricing for train tickets.

(d) Bundling vs. a two-part tariff — 15%

	Bundling	Two-part tariff
What it is	Selling two or more distinct products together as a single package, often at a combined price lower than buying each separately.	A pricing structure for a single product/service consisting of a fixed fee (paid regardless of usage) plus a per-unit price for actual usage.
Mechanism for extracting surplus	Exploits differences in how much different consumers value each individual component — a consumer who values one component highly and another low, bundled with a consumer who values them the opposite way, can both be charged a similar bundle price that extracts more total surplus than pricing each component separately.	The fixed fee extracts a portion of consumer surplus up front (ideally set close to the surplus of the lowest-value consumer the firm still wants to serve), while the per-unit price can be set closer to marginal cost to maximise the quantity consumed (and hence the surplus available to extract via the fixed fee).
Example	Microsoft Office suite (Word, Excel, PowerPoint sold as one package); cable TV channel packages; fast-food value meals (burger+fries+drink).	A gym membership (monthly fee + optional per-class fee); Costco (annual membership fee + per-item prices); a mobile phone plan (line rental fee + per-GB data charges).

> **The common thread:** both are methods of extracting more total consumer surplus than a single, simple per-unit price would allow — but bundling does this by combining multiple different products, while a two-part tariff does it by splitting the pricing of a single product into a fixed and a variable component.

Mark Allocation — Part (a) (20% + 10%)

Tier	Marks	What separates this tier
Full marks	Correct HHI calculation (2,204) with all seven terms shown, and correct interpretation using the standard concentration thresholds.	
Partial	Correct calculation but limited interpretation, or a minor arithmetic slip.	
Minimal	Incorrect HHI formula or no meaningful interpretation.	

Mark Allocation — Part (b) (10%)

Tier	Marks	What separates this tier
8–10%	All three conditions correctly identified and explained.	
4–7%	Two of three conditions identified.	
0–3%	One or none identified.	

Mark Allocation — Part (c) (3 × 15%)

Tier	Marks	What separates this tier
12–15% per degree	Correct diagram, correct distinguishing explanation, and a relevant, correctly-classified example.	
6–11% per degree	Correct general description but diagram or example is missing/weak.	
0–5% per degree	Incorrect or minimal treatment.	

Mark Allocation — Part (d) (15%)

Tier	Marks	What separates this tier
12–15%	Correctly distinguishes bundling (multiple products) from a two-part tariff (single product, fixed+variable pricing), with a correct example of each.	
6–11%	Correct examples but the underlying distinction is not clearly articulated.	
0–5%	Incorrect or minimal treatment.	

Question B1 Total: 100%

Section B · Question 2

A Perfectly Competitive Industry & Firm · 20+20+30+30%

B2

Table B2: Industry and Firm Data on a Perfectly Competitive Firm

Industry: $Q_s = 3000 + 200P$; $Q_d = 13{,}500 - 500P$

Firm: Fixed Cost (FC) = 50; Marginal Cost (MC) = $15 - 4Q + 3Q^2/10$; Average Variable Cost (AVC) = $15 - 2Q + Q^2/10$

(where Q = Quantity, Q_s = quantity supplied, Q_d = quantity demanded, P = price)

B2(a) **20%**

Calculate the equilibrium price and output in the industry in Table B2.

B2(b) **20%**

Calculate the profit maximising level of output for the firm in Table B2.

B2(c) **30%**

Calculate the profit (π) earned by the firm in Table B2 at the profit maximising level of output.

B2(d) **30%**

Provide an illustration of the three stages of production and clearly describe the interactions between the Total Product (TP), Marginal Product (MP), Average Product (AP) and elasticity.

FULL MODEL ANSWER

(a) Industry equilibrium — 20%

Step	Working
Set $Q_s = Q_d$	$3000 + 200P = 13{,}500 - 500P$
Collect terms	$700P = 10{,}500$
Solve for P*	**P* = €15**
Substitute into either function for Q*	**Q* = 3000 + 200(15) = 3000 + 3000 = 6,000**

(b) The firm's profit-maximising output — 20%

As a price-taking firm in a perfectly competitive market, the firm treats the market price (P*=€15 from part (a)) as given, and profit is maximised where **P = MC**.

Step	Working
Set MC = P	$15 - 4Q + 0.3Q^2 = 15$
Simplify	$-4Q + 0.3Q^2 = 0 \Rightarrow Q(0.3Q - 4) = 0$
Solve	Q = 0 (rejected, not a profit-maximising interior solution) or Q* = 4/0.3
Profit-maximising output	**Q* = 13.33 (to 2 d.p.)**

Checking the shutdown condition: the firm should only produce if $P \geq AVC$ at this output. AVC at $Q=13.33$ is $15-2(13.33)+(13.33)^2/10 \approx$ €6.10, well below P=€15 — confirming the firm should indeed produce at $Q^*=13.33$ rather than shut down.

(c) Profit at the profit-maximising output — 30%

Step	Working
Variable cost, $VC = AVC \times Q$	$VC = (15 - 2Q + Q^2/10) \times Q = 15Q - 2Q^2 + 0.1Q^3$
Check: $d(VC)/dQ$ should equal the given MC	$d(VC)/dQ = 15 - 4Q + 0.3Q^2$ — matches the given MC exactly ✓
Total cost, $TC = FC + VC$	$TC = 50 + 15Q - 2Q^2 + 0.1Q^3$
Total revenue, $TR = P \times Q$	$TR = 15Q$
Profit, $\pi = TR - TC$	**$\pi = 15Q - (50 + 15Q - 2Q^2 + 0.1Q^3) = -0.1Q^3 + 2Q^2 - 50$**
Evaluate at $Q^*=13.33$	$\pi^* = -0.1(13.33)^3 + 2(13.33)^2 - 50$
Profit at the profit-maximising output	**$\pi^* \approx$ €68.52**

(d) The three stages of production, including elasticity — 30%

The three stages of production

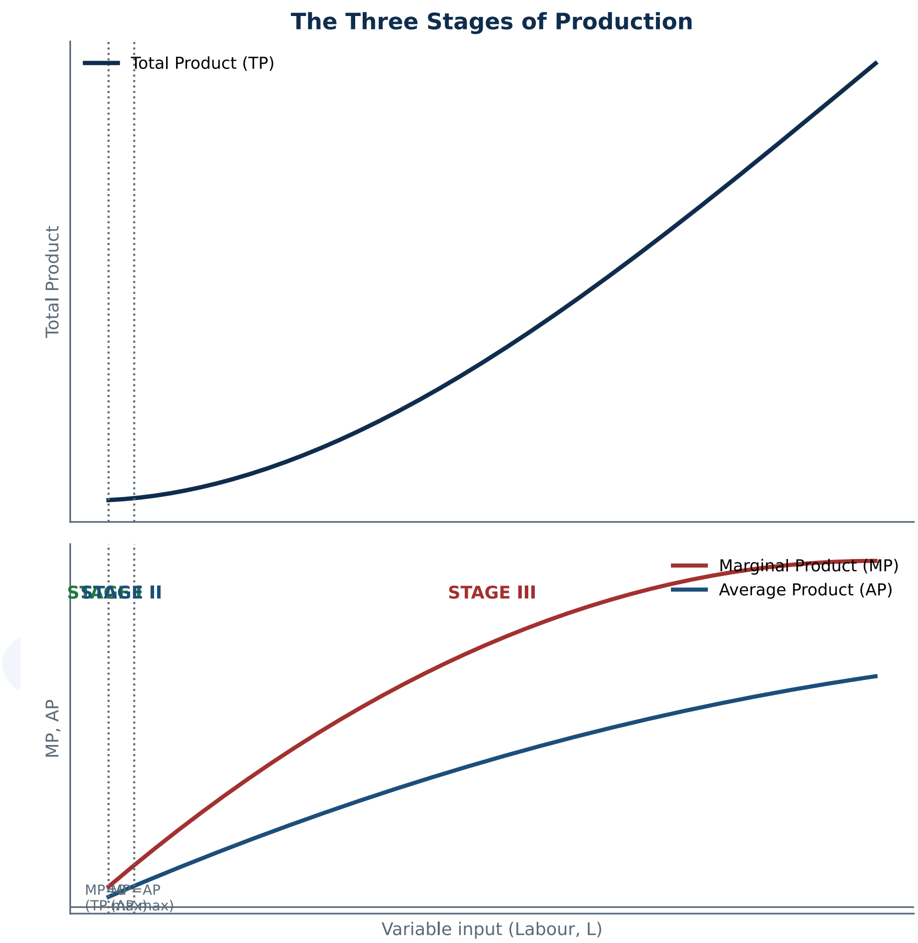

Fig. — Total, Marginal, and Average Product across the three stages of production

Stage	Boundaries	What's happening
Stage I	From zero input up to where MP = AP (AP at its maximum)	MP is rising and exceeds AP, pulling AP up; each additional unit of the variable input adds more to output than the average unit before it — increasing returns.
Stage II	From MP=AP up to where MP = 0 (TP at its maximum)	MP is falling (diminishing marginal returns have set in) but is still positive and below AP, so AP is falling too; TP is still rising, just at a decreasing rate. **This is the economically rational stage in which to operate** — the variable input is still adding to output, and the firm can choose the specific point within this stage using marginal analysis (MR=MC).
Stage III	Beyond MP = 0	MP is negative — adding more of the variable input actually reduces total output (e.g. overcrowding a fixed amount of capital/land with too much labour). No rational firm would operate here, since less input would produce more output.

> **The AP/MP relationship (the key interaction the diagram illustrates):** whenever MP is above AP, AP is rising (Stage I); whenever MP is below AP, AP is falling (Stages II-III); MP crosses AP exactly at AP's maximum point — the same mathematical relationship as between a marginal and average curve anywhere in economics (e.g. marginal and average cost).

> **Why Stage II is where a profit-maximising firm operates:** in Stage I, the marginal product of the variable input is still rising, meaning the firm isn't yet using its fixed inputs intensively enough — it should keep adding the variable input. In Stage III, marginal product is negative, so the firm is actively using too much of the variable input. Only in Stage II does adding the variable input contribute positively but at a diminishing rate, which is exactly where the marginal-analysis (MR=MC) condition for profit maximisation can select a genuine interior optimum.

> **Bringing in the elasticity of production:** the elasticity of production, E_p = MP/AP, gives a compact way to characterise the three stages: in **Stage I**, MP>AP so $E_p>1$ (output is growing proportionally faster than the variable input — a 1% rise in labour raises output by more than 1%); in **Stage II**, MP<AP but MP>0, so $0<E_p<1$ (output still rises with more input, but proportionally more slowly); in **Stage III**, MP<0, so $E_p<0$ (output actually falls as more of the variable input is added). A profit-maximising firm always operates in Stage II, where $0<E_p<1$, since Stage I still has room to profitably expand and Stage III is unambiguously wasteful.

Mark Allocation — Part (a) (20%)

Tier	Marks	What separates this tier
16–20%	Correctly sets Qs=Qd, solves for P*=15 and Q*=6,000, workings shown.	
8–15%	Correct method with an arithmetic slip carried through consistently.	
0–7%	Incorrect equilibrium condition or method.	

Mark Allocation — Part (b) (20%)

Tier	Marks	What separates this tier
16–20%	Correctly identifies P=MC as the profit-maximising condition for a price-taking firm, uses P*=15 from part (a), and correctly solves for Q*=13.33.	
8–15%	Correct method with an arithmetic error, or the P=MC condition not explicitly justified.	
0–7%	Incorrect method (e.g. MR=MC used without recognising P=MR for a price taker, or MC set equal to the wrong value).	

Mark Allocation — Part (c) (30%)

Tier	Marks	What separates this tier
22–30%	Correctly derives VC from AVC, correctly constructs TC and TR, correctly derives the profit function, and correctly evaluates it at $Q^*=13.33$ for $\pi^*\approx€68.52$.	
12–21%	Correct method with an own-figure error carried through from part (b), or a computational slip.	
0–11%	Incorrect construction of TC/TR/profit.	

Mark Allocation — Part (d) (30%)

Tier	Marks	What separates this tier
22–30%	Correct three-stage diagram with correct TP/MP/AP interactions AND a correct discussion of the elasticity of production (E_p=MP/AP) characterising each stage.	
12–21%	Correct diagram and TP/MP/AP interactions but the elasticity discussion is missing or incorrect.	
0–11%	Minimal or incorrect treatment.	

Question B2 Total: 100%